

DISCRETE-TIME SYSTEM STRUCTURES

ENSC 429 – Digital Signal Processing

School of Engineering Science
Simon Fraser University
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DIFFERENCE EQUATIONS FOR RATIONAL SYSTEM FUNCTIONS

- Rational system functions can be represented by difference equations

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1} + \dots + b_M z^{-M}}{a_0 + a_1 z^{-1} + \dots + a_N z^{-N}}$$

$$Y(z)[a_0 + a_1 z^{-1} + \dots + a_N z^{-N}] = X(z)[b_0 + b_1 z^{-1} + \dots + b_M z^{-M}]$$

$$a_0 Y(z) + a_1 Y(z)z^{-1} + \dots + a_N Y(z)z^{-N} = b_0 X(z) + b_1 X(z)z^{-1} + \dots + b_M X(z)z^{-M}$$

$$a_0 y[n] + a_1 y[n-1] + \dots + a_N y[n-N] = b_0 x[n] + b_1 x[n-1] + \dots + b_M x[n-M]$$

- Divide by a_0

$$y[n] + \frac{a_1}{a_0} y[n-1] + \dots + \frac{a_N}{a_0} y[n-N] = \frac{b_0}{a_0} x[n] + \frac{b_1}{a_0} x[n-1] + \dots + \frac{b_M}{a_0} x[n-M]$$

Minus sign!

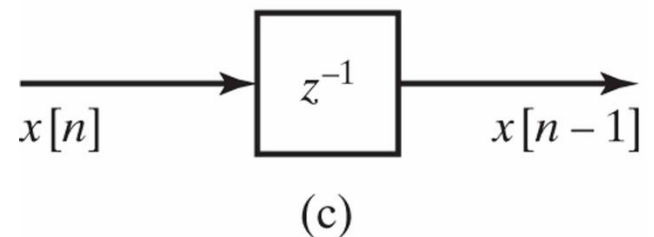
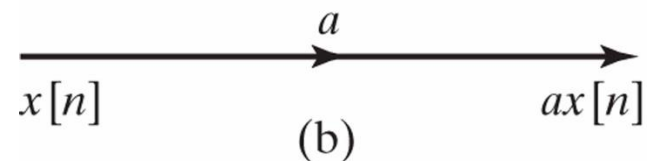
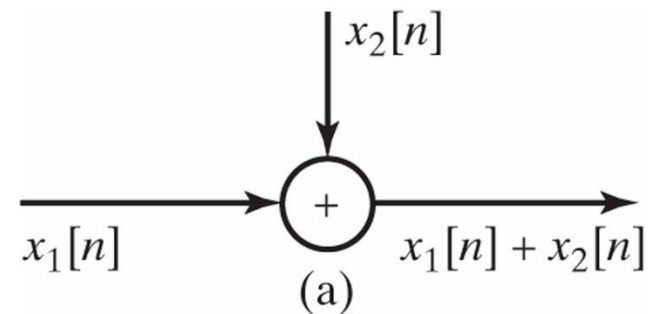
- Let $a'_i = -a_i/a_0$ and $b'_i = b_i/a_0$, then

$$y[n] = a'_1 y[n-1] + \dots + a'_N y[n-N] + b'_0 x[n] + b'_1 x[n-1] + \dots + b'_M x[n-M]$$

- This shows how $y[n]$ can be computed based on previous outputs and inputs

BLOCK DIAGRAMS FOR DIFFERENCE EQUATIONS

- How can a linear constant-coefficient difference equation (LCCDE) be implemented?
- It turns out we only need three basic elements:
 - a) Addition of two elements
 - b) Multiplication by a constant
 - c) Unit delay (delay by one sample)



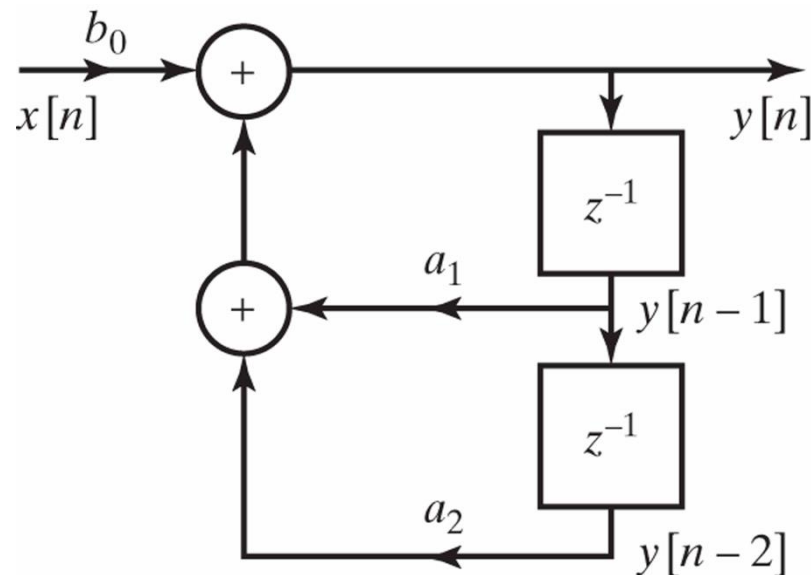
BLOCK DIAGRAMS FOR DIFFERENCE EQUATIONS

- Example:

$$H(z) = \frac{b_0}{1 - a_1 z^{-1} - a_2 z^{-2}}$$

- Difference equation ([see notes](#)):

$$y[n] = a_1 y[n-1] + a_2 y[n-2] + b_0 x[n]$$



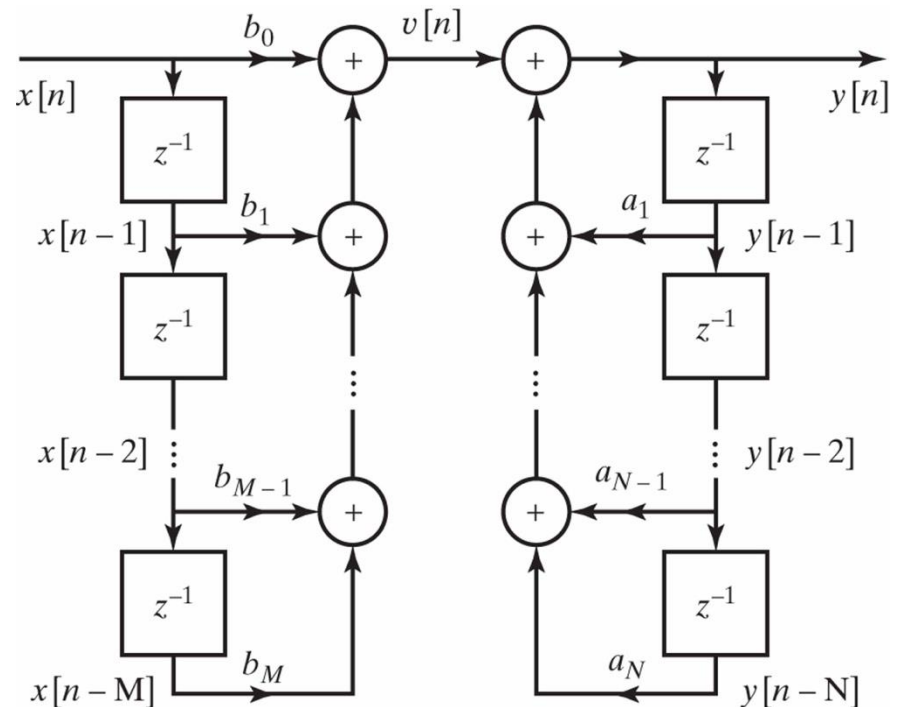
BLOCK DIAGRAMS FOR DIFFERENCE EQUATIONS

- Example:

$$H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{1 - \sum_{k=1}^N a_k z^{-k}}$$

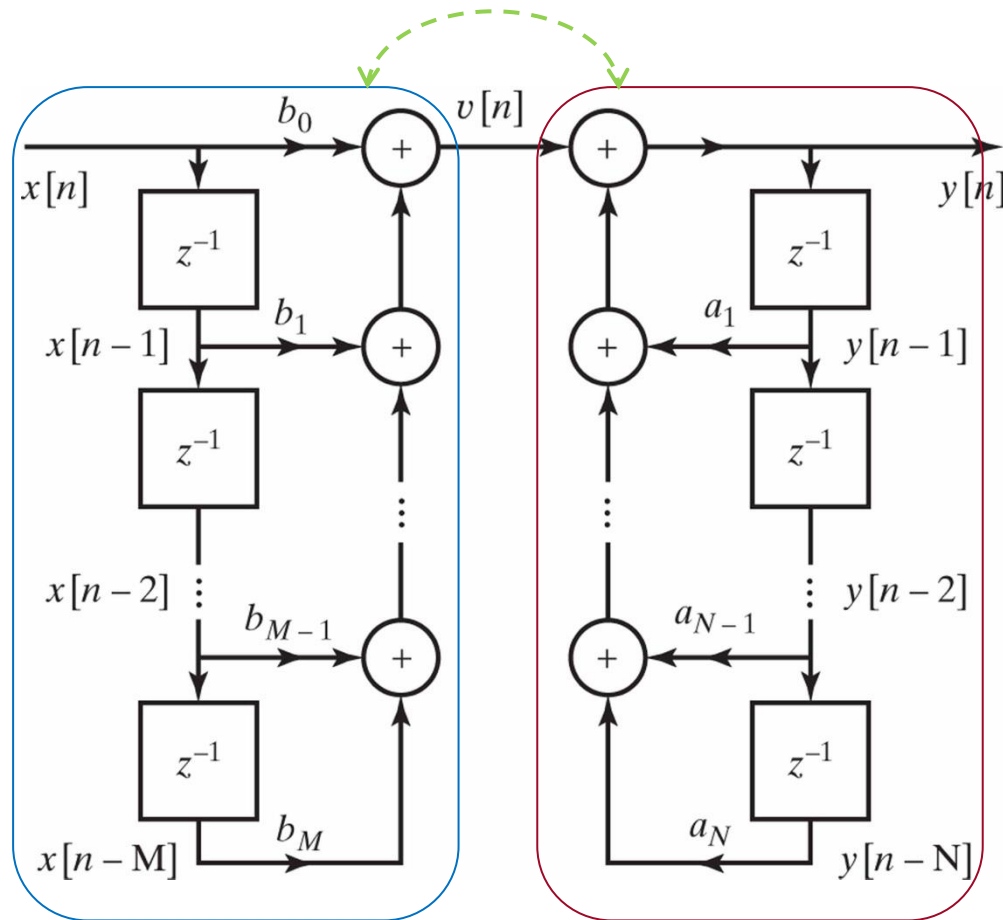
- Difference equation ([see notes](#)):

$$y[n] = \sum_{k=1}^N a_k y[n-k] + \underbrace{\sum_{k=0}^M b_k x[n-k]}_{v[n]}$$



Direct form I

BLOCK DIAGRAMS FOR DIFFERENCE EQUATIONS

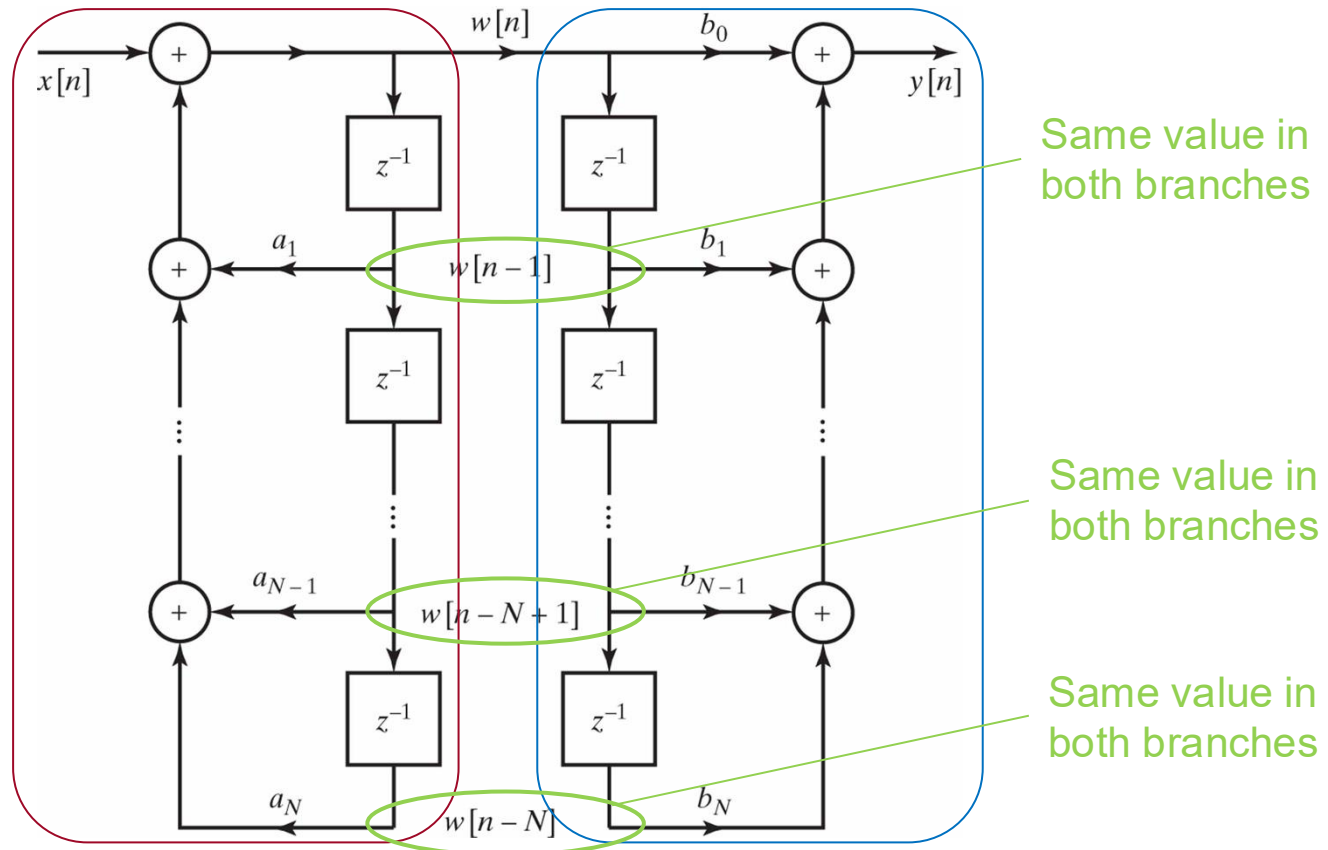


LTI systems can interchange the order

$$H_1(z) = \sum_{k=0}^M b_k z^{-k}$$

$$H_2(z) = \frac{1}{1 - \sum_{k=1}^N a_k z^{-k}}$$

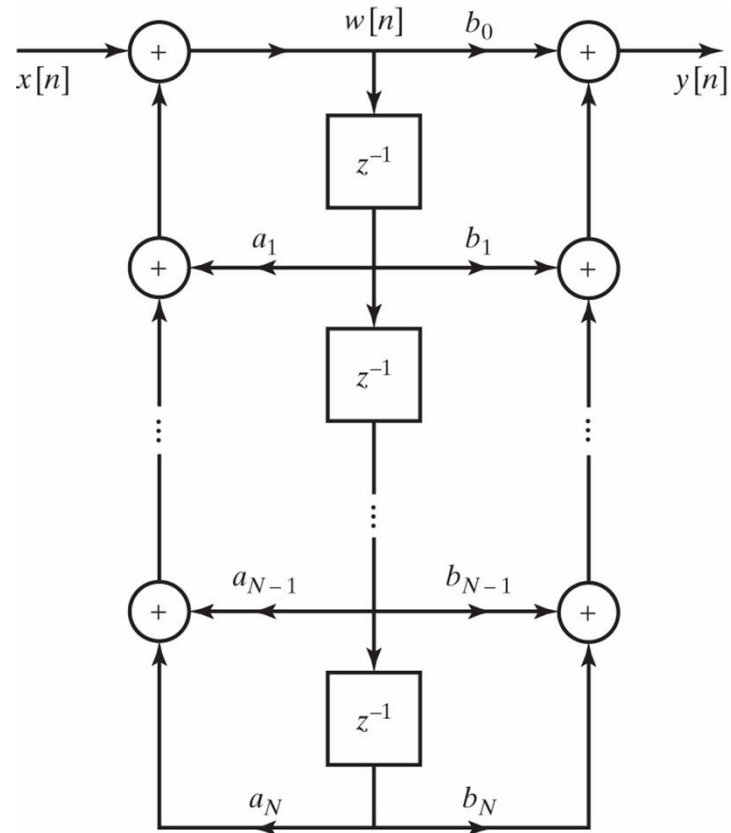
BLOCK DIAGRAMS FOR DIFFERENCE EQUATIONS



$$H_2(z) = \frac{1}{1 - \sum_{k=1}^N a_k z^{-k}}$$

$$H_1(z) = \sum_{k=0}^M b_k z^{-k}$$

BLOCK DIAGRAMS FOR DIFFERENCE EQUATIONS

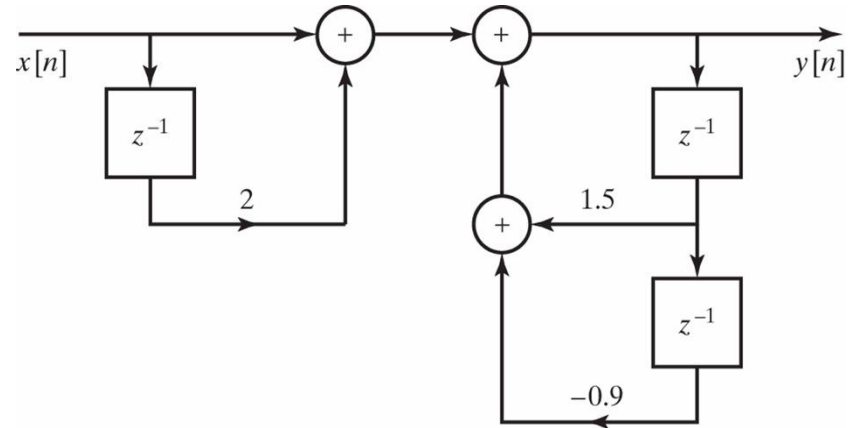


Direct form II or canonic direct form

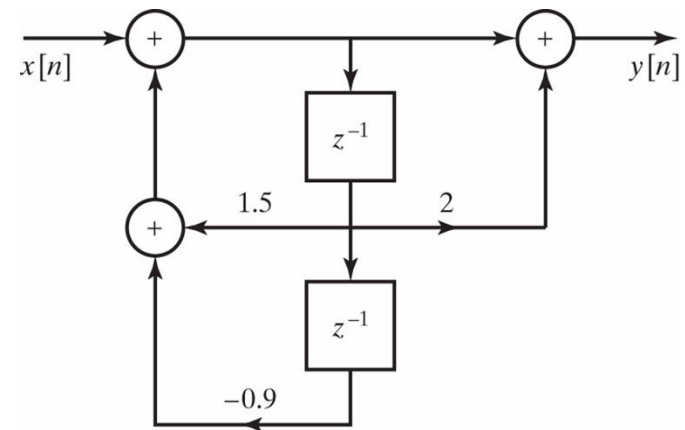
BLOCK DIAGRAMS FOR DIFFERENCE EQUATIONS

Example:

$$H(z) = \frac{1 + 2z^{-1}}{1 - 1.5z^{-1} + 0.9z^{-2}}$$



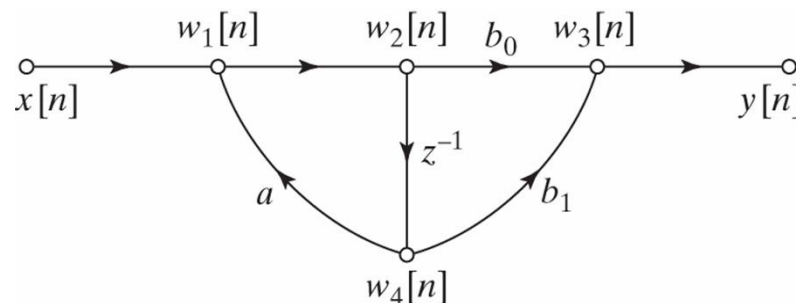
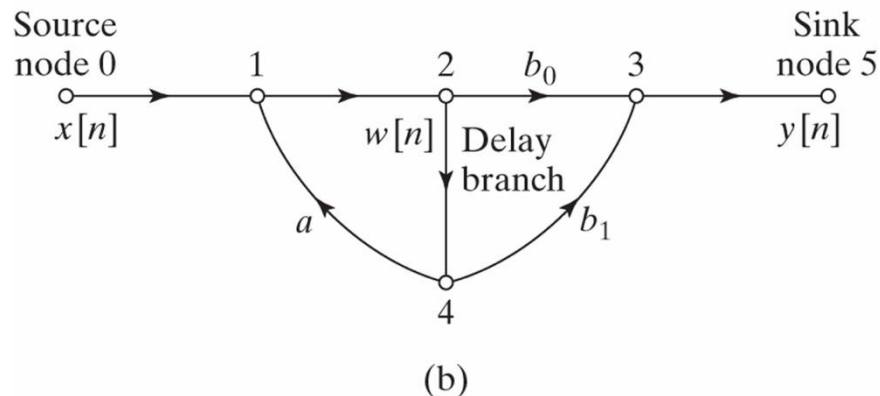
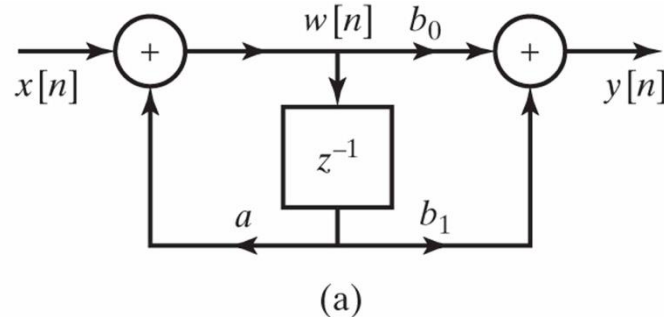
Direct form I



Direct form II

SIGNAL FLOW GRAPHS

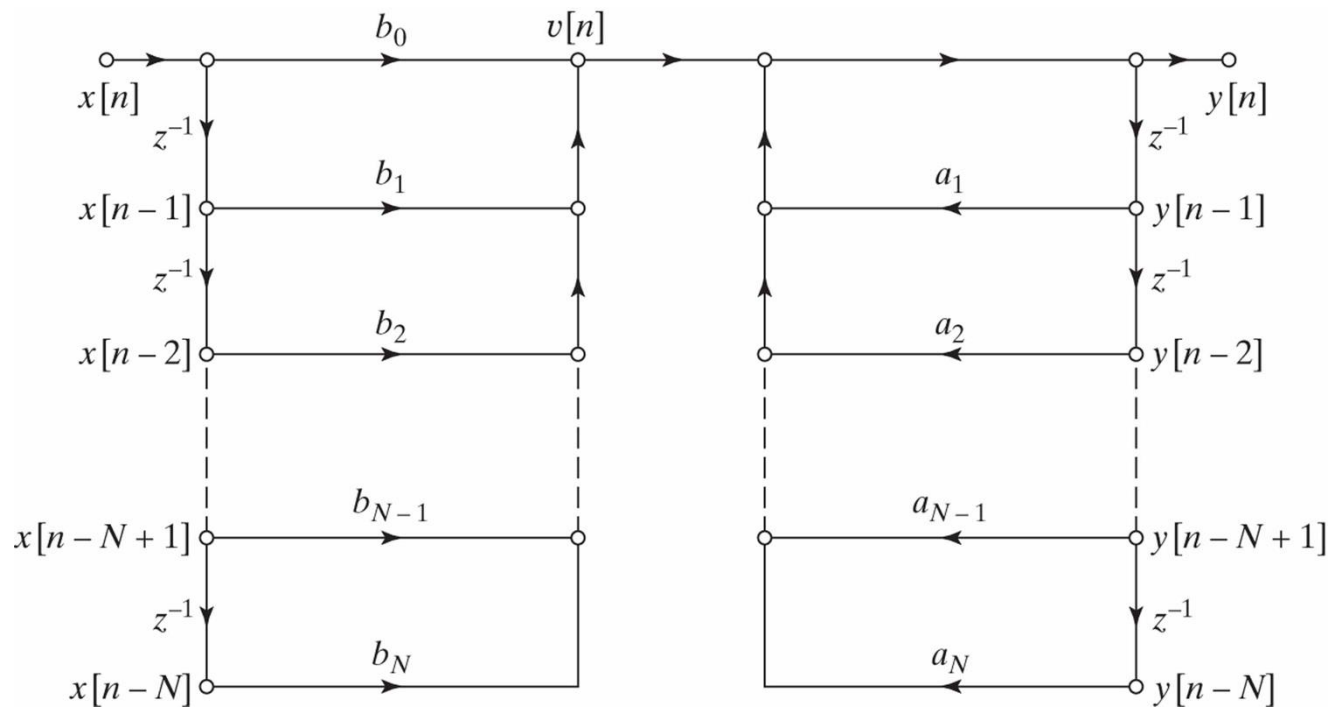
- Signal flow graphs are a simpler alternative to block diagrams
- Gains are still written next to each branch
- Delay is treated as a gain of z^{-1} because this is the z-transform of the unit delay



STRUCTURES FOR IIR SYSTEMS

Direct form I realization of

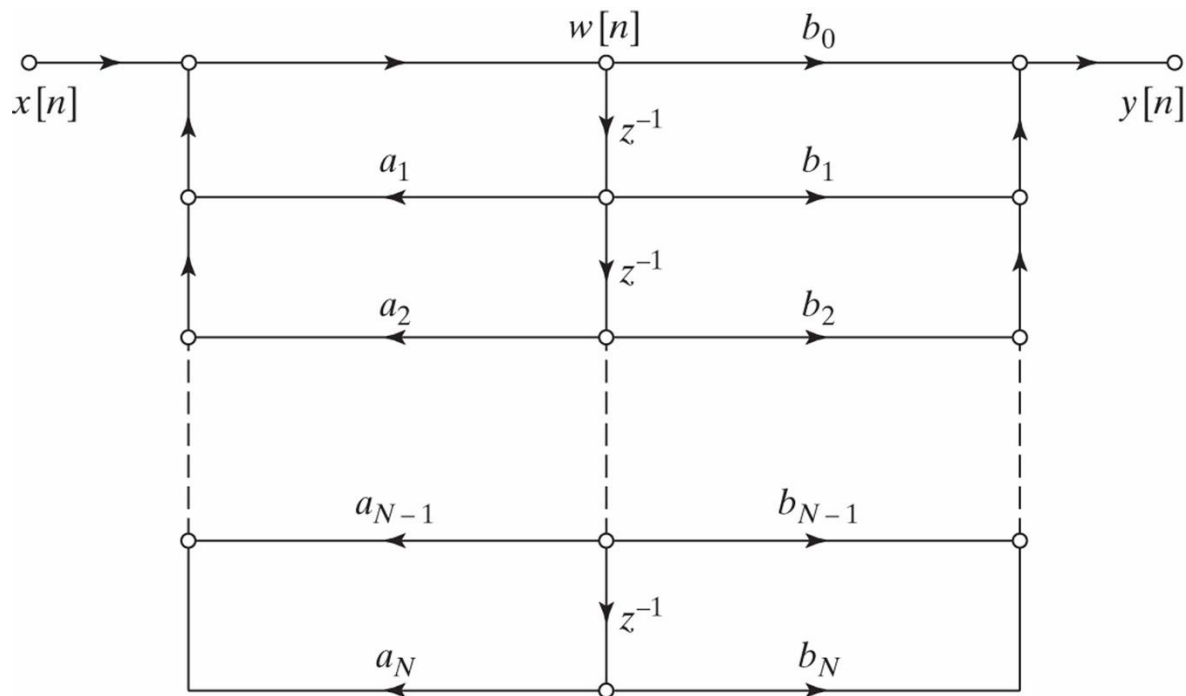
$$H(z) = \frac{\sum_{k=0}^N b_k z^{-k}}{1 - \sum_{k=1}^N a_k z^{-k}}$$



STRUCTURES FOR IIR SYSTEMS

Direct form II realization of

$$H(z) = \frac{\sum_{k=0}^N b_k z^{-k}}{1 - \sum_{k=1}^N a_k z^{-k}}$$

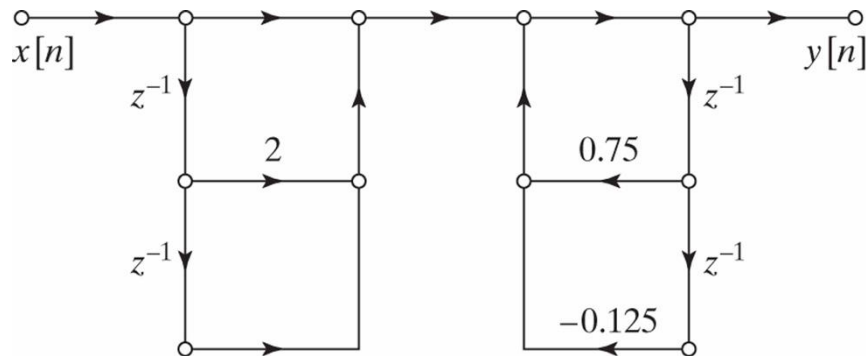


STRUCTURES FOR IIR SYSTEMS

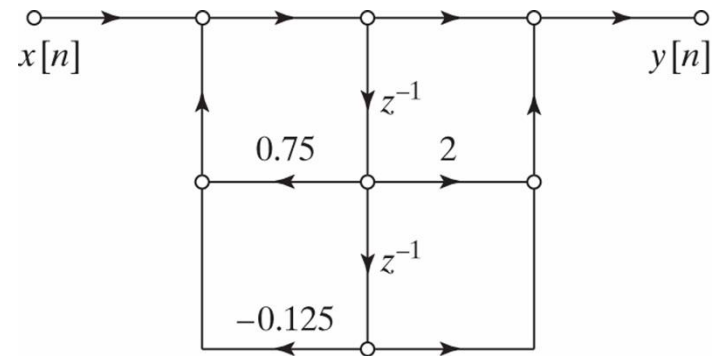
Example:

$$H(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - 0.75z^{-1} + 0.125z^{-2}}$$

Direct form I



Direct form II



STRUCTURES FOR IIR SYSTEMS

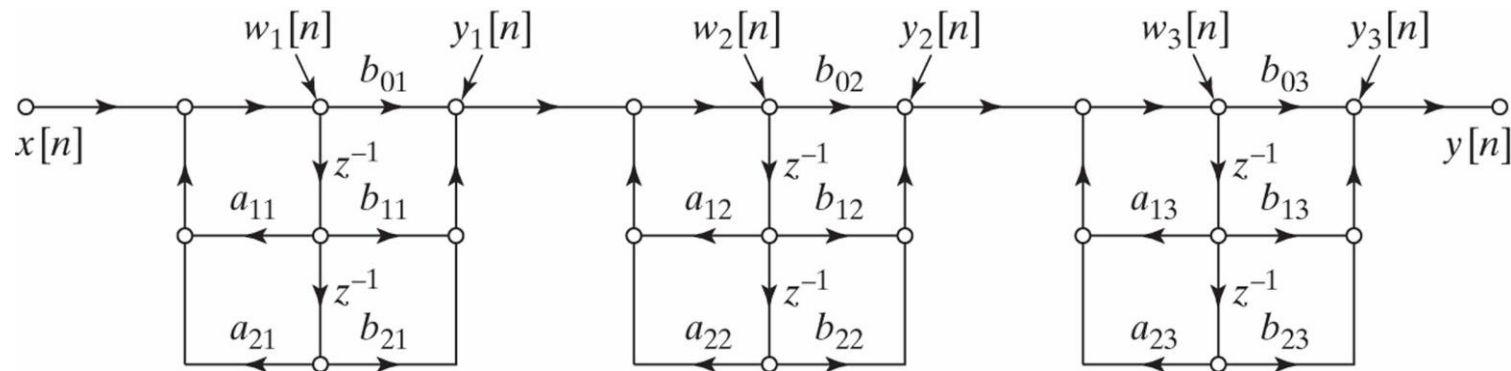
- Cascade form:** factor the numerator and denominator, then write as product of 1st and 2nd order sections

$$H(z) = A \frac{\prod_{k=1}^{M_1} (1 - f_k z^{-1}) \prod_{k=1}^{M_2} (1 - g_k z^{-1})(1 - g_k^* z^{-1})}{\prod_{k=1}^{N_1} (1 - c_k z^{-1}) \prod_{k=1}^{N_2} (1 - d_k z^{-1})(1 - d_k^* z^{-1})}$$

$$= \prod_{k=1}^{N_s} \frac{b_{0k} + b_{1k} z^{-1} + b_{2k} z^{-2}}{1 - a_{1k} z^{-1} - a_{2k} z^{-2}}$$

Zero if 1st order section

- Example: cascade of three 2nd order sections

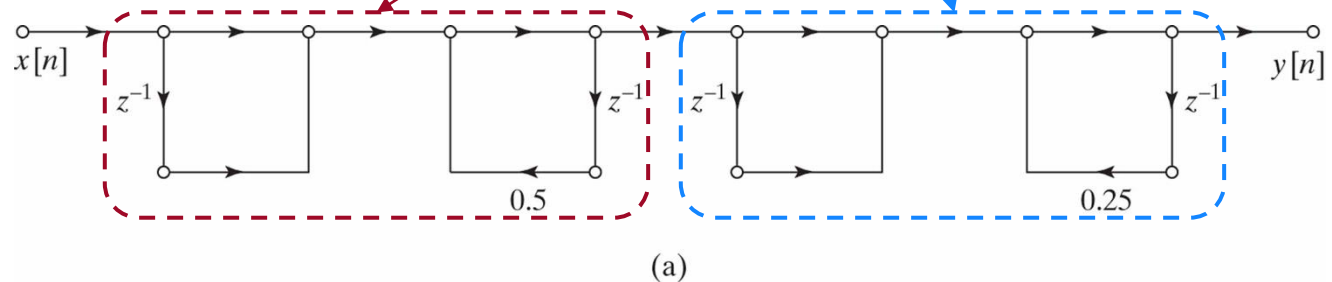


STRUCTURES FOR IIR SYSTEMS

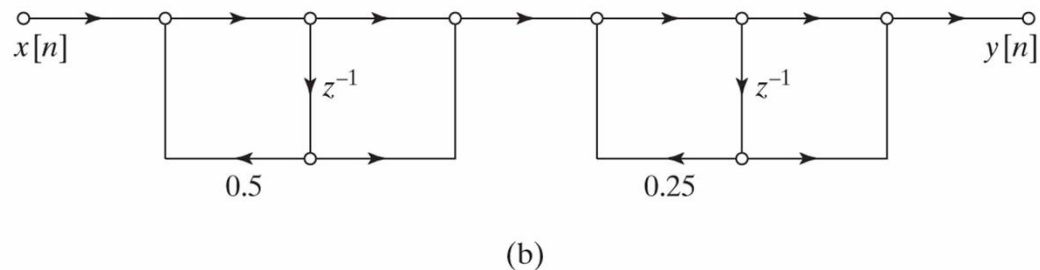
Example: same $H(z)$ as on slide 13

$$H(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - 0.75z^{-1} + 0.125z^{-2}} = \frac{(1 + z^{-1})(1 + z^{-1})}{(1 - 0.5z^{-1})(1 - 0.25z^{-1})}$$

Cascade of direct form I sections



Cascade of direct form II sections



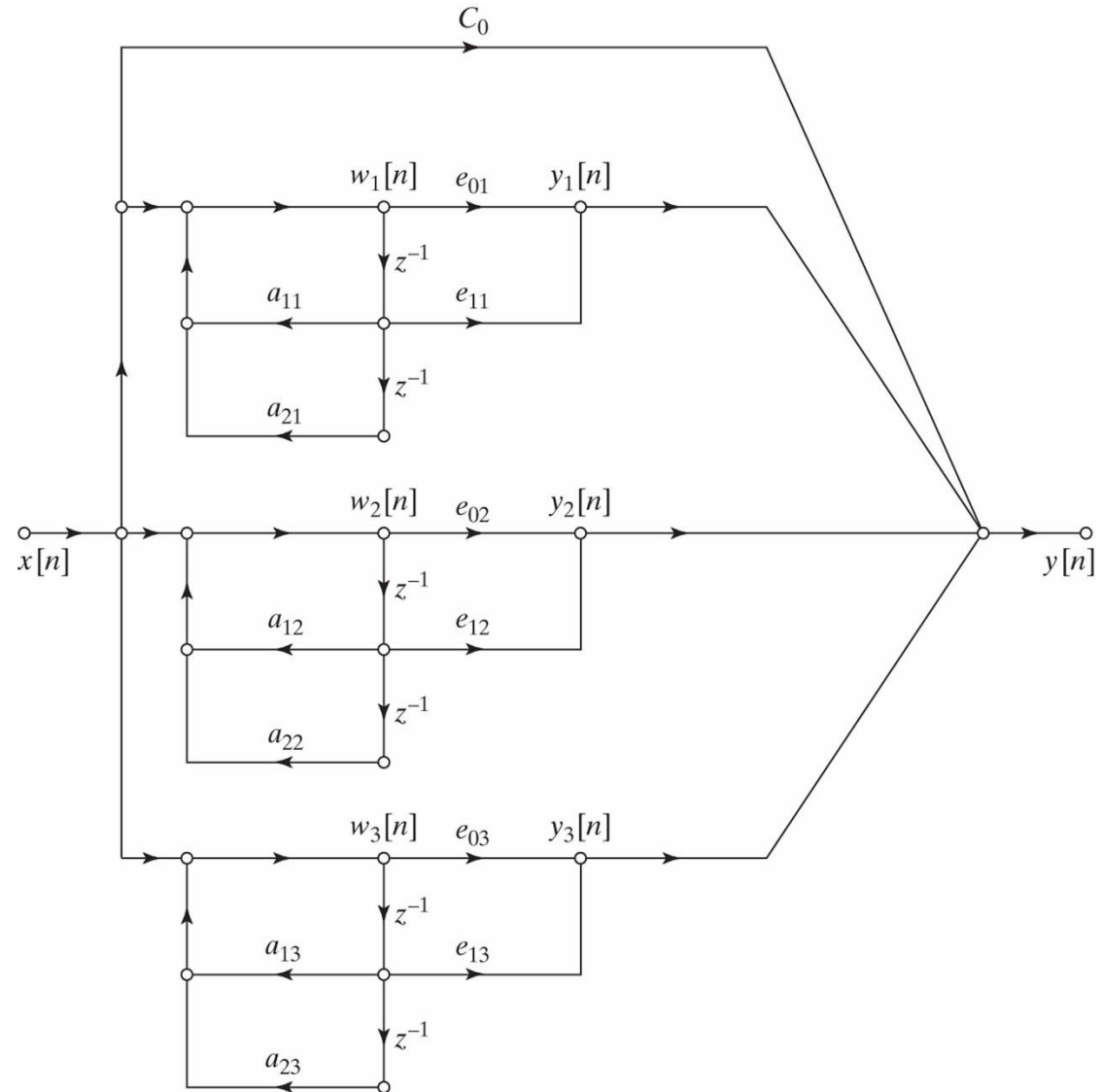
STRUCTURES FOR IIR SYSTEMS

- **Parallel form:** using partial fraction expansion, decompose $H(z)$, then group poles into 2nd order sections

$$H(z) = \sum_{k=0}^{N_p} C_k z^{-k} + \sum_{k=1}^{N_1} \frac{A_k}{1 - c_k z^{-1}} + \sum_{k=1}^{N_2} \frac{B_k(1 - e_k z^{-1})}{(1 - d_k z^{-1})(1 - d_k^* z^{-1})}$$
$$= \sum_{k=0}^{N_p} C_k z^{-k} + \sum_{k=1}^{N_s} \frac{e_{0k} + e_{1k} z^{-1}}{1 - a_{1k} z^{-1} - a_{2k} z^{-2}}$$

STRUCTURES FOR IIR SYSTEMS

Example: parallel form
consisting of three 2nd
order sections

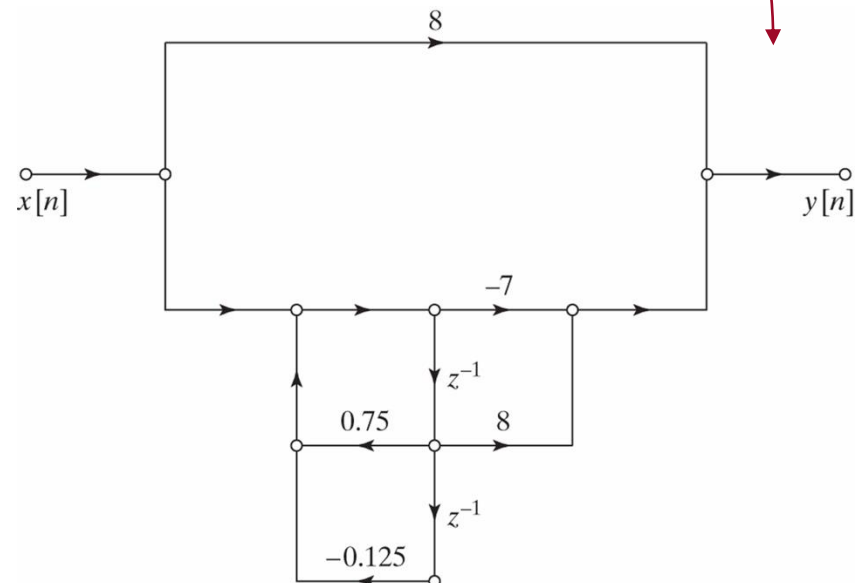
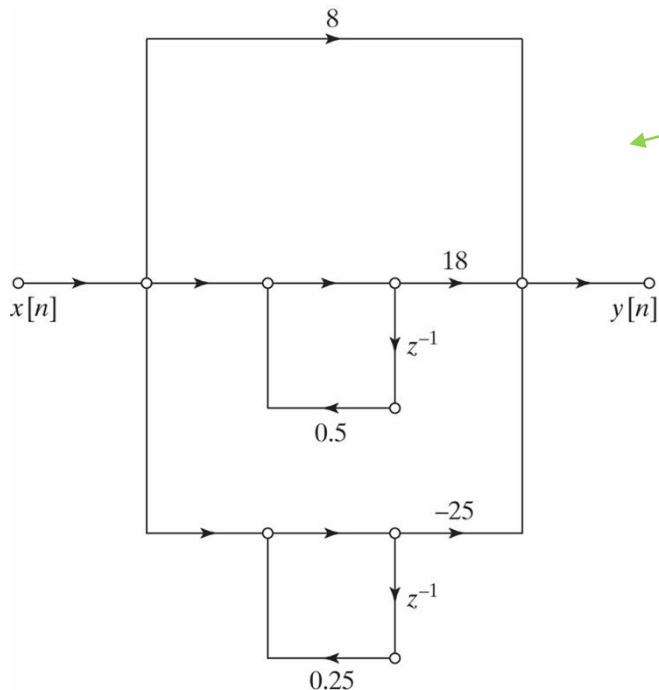


STRUCTURES FOR IIR SYSTEMS

Example: same $H(z)$ as on slide 13 ([see notes](#))

$$H(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - 0.75z^{-1} + 0.125z^{-2}} = 8 + \frac{-7 + 8z^{-1}}{1 - 0.75z^{-1} + 0.125z^{-2}}$$

$$= 8 + \frac{18}{1 - 0.5z^{-1}} - \frac{25}{1 - 0.25z^{-1}}$$



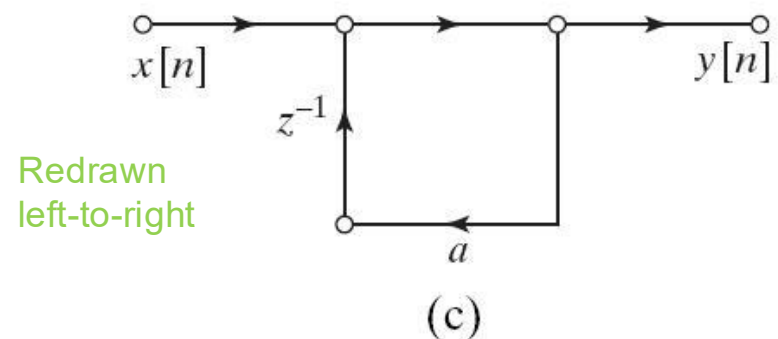
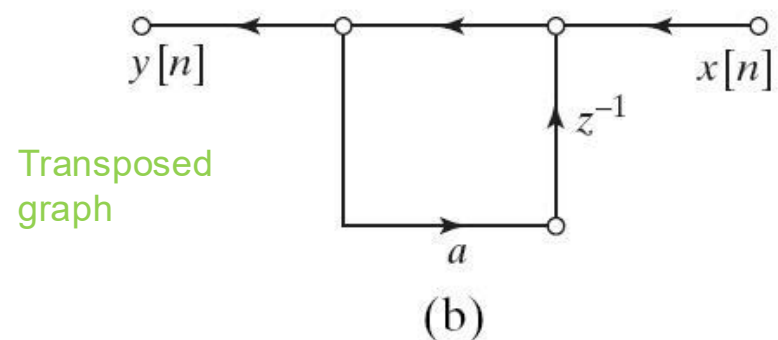
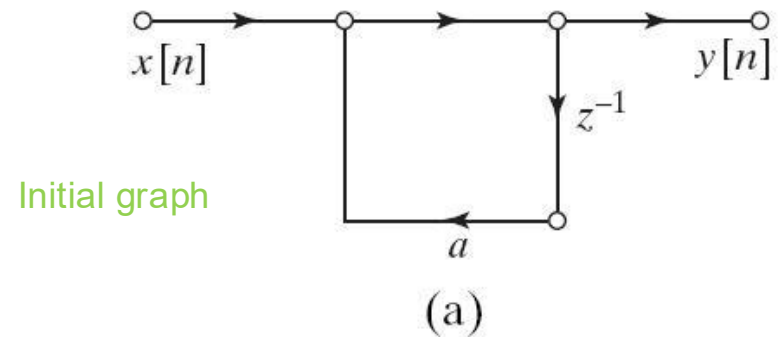
TRANPOSED FORMS

- **Transposing** a signal flow graph means reversing the direction of each branch, and exchanging the roles of input and output
- It can be shown that transposing a signal flow graph does not change the system function
- This gives us yet another way to implement a given system function
- Example:

$$H(z) = \frac{1}{1 - az^{-1}}$$

- Difference equation:

$$y[n] = ay[n - 1] + x[n]$$



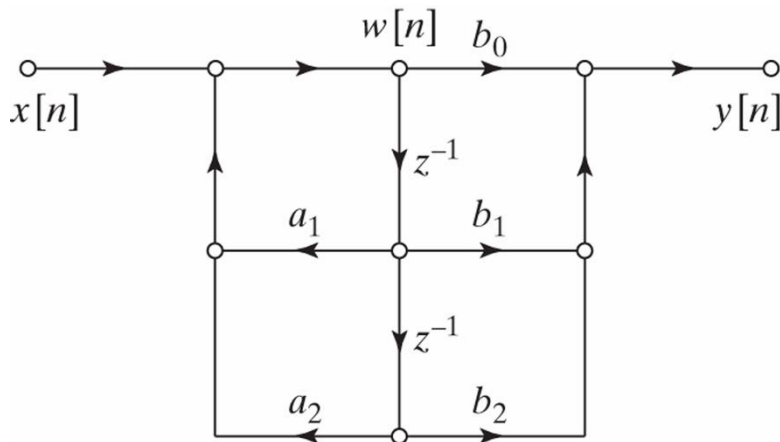
TRANPOSED FORMS

Example:

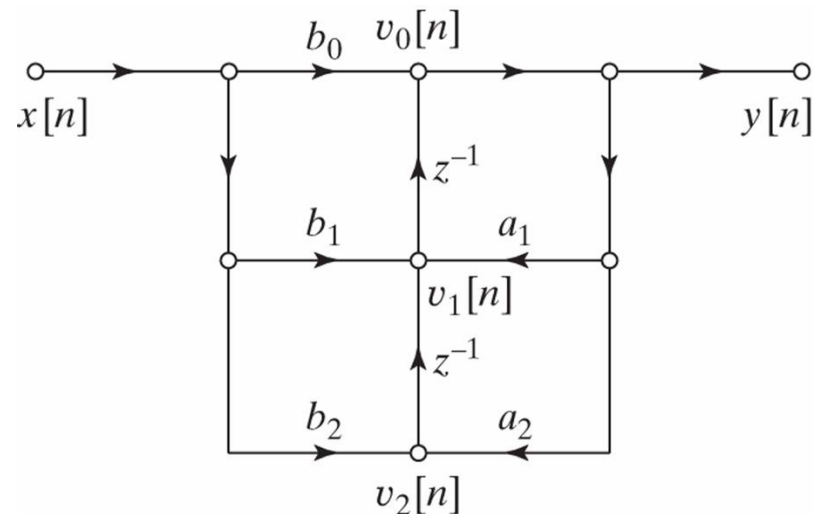
$$H(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{1 - a_1 z^{-1} - a_2 z^{-2}}$$

(see notes)

Direct form II

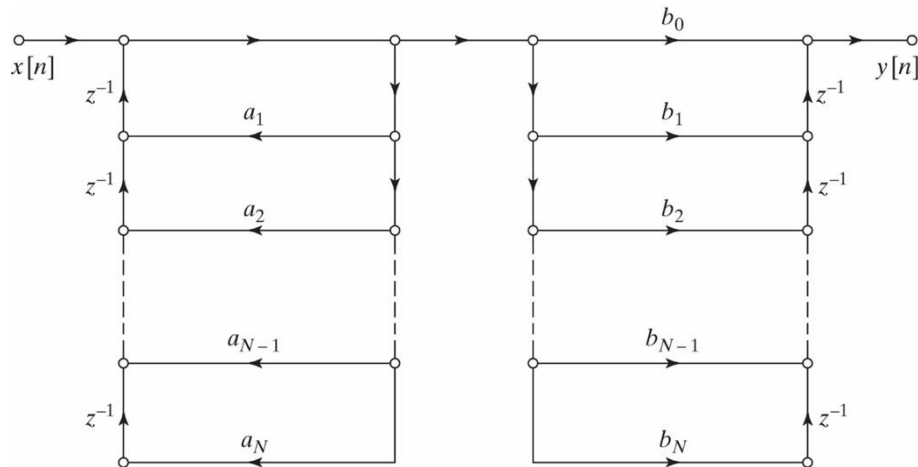


Transposed direct form II

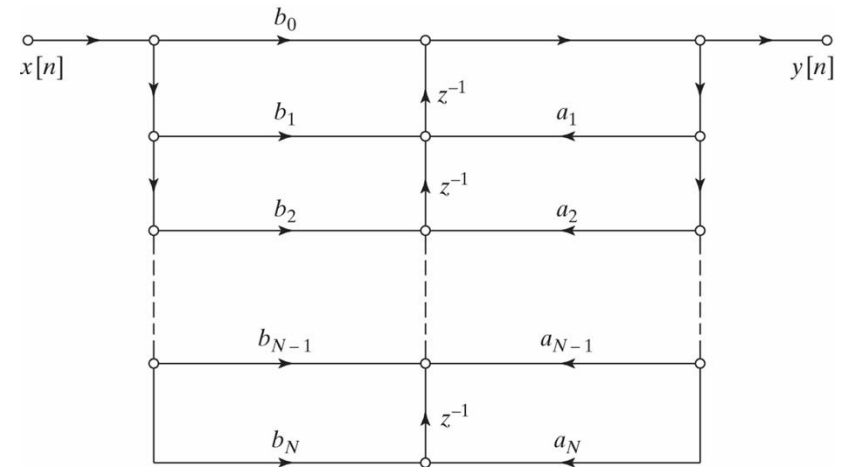


TRANPOSED FORMS

General transposed direct form I



General transposed direct form II



STRUCTURES FOR FIR SYSTEMS

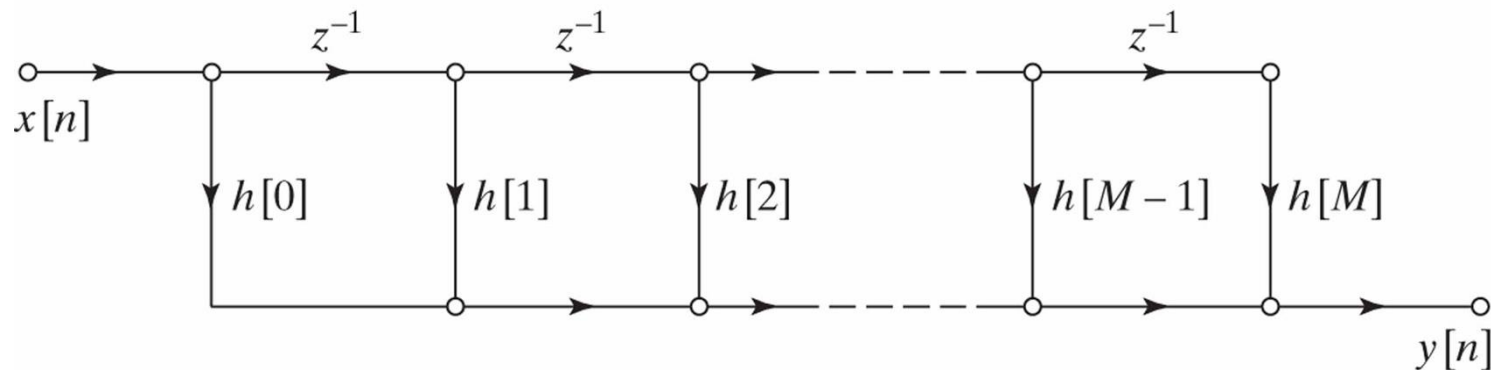
- Causal FIR impulse response:

$$h[n] = \begin{cases} b_n & n = 0, 1, \dots, M \\ 0 & \text{else} \end{cases}$$

- For the input signal $x[n]$, the output is

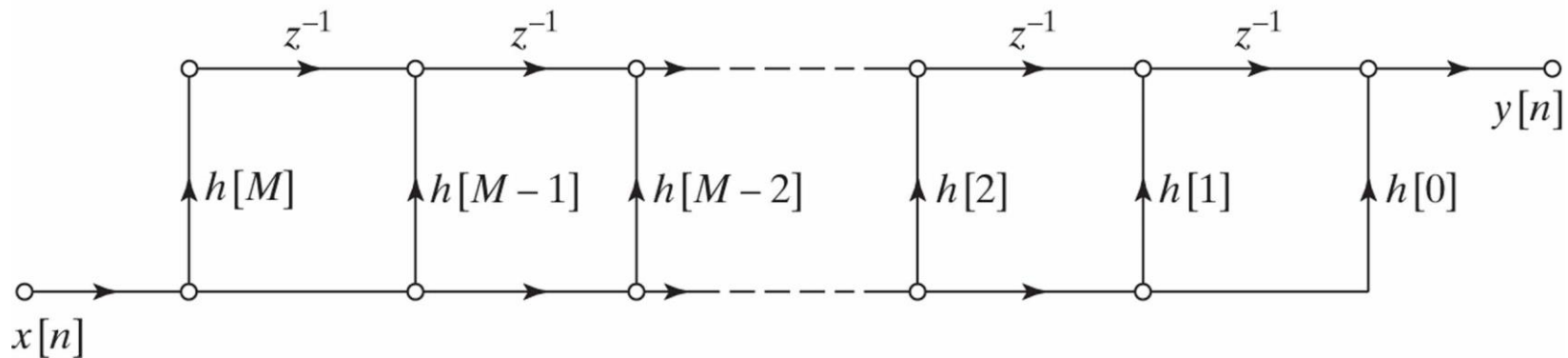
$$y[n] = h[n] * x[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k] = \sum_{k=0}^M b_k x[n-k]$$

- **Direct form** realization:



STRUCTURES FOR FIR SYSTEMS

- Of course, we can also have the **transposed direct form** realization:

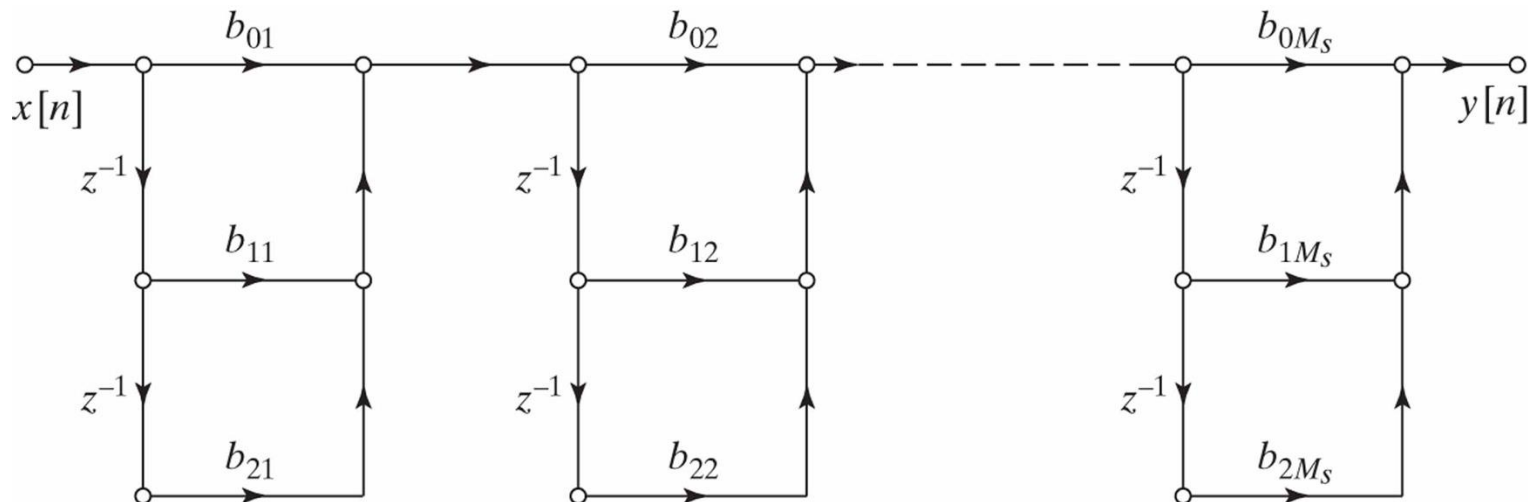


STRUCTURES FOR FIR SYSTEMS

- **Cascade forms** can be obtained by factoring the system function polynomial
- For example, if the system function (of even degree) is factored into 2nd-order factors

$$H(z) = \sum_{n=0}^M h[n]z^{-n} = \prod_{k=1}^{M_s} (b_{0k} + b_{1k}z^{-1} + b_{2k}z^{-2})$$

then the following cascade form realization is obtained:



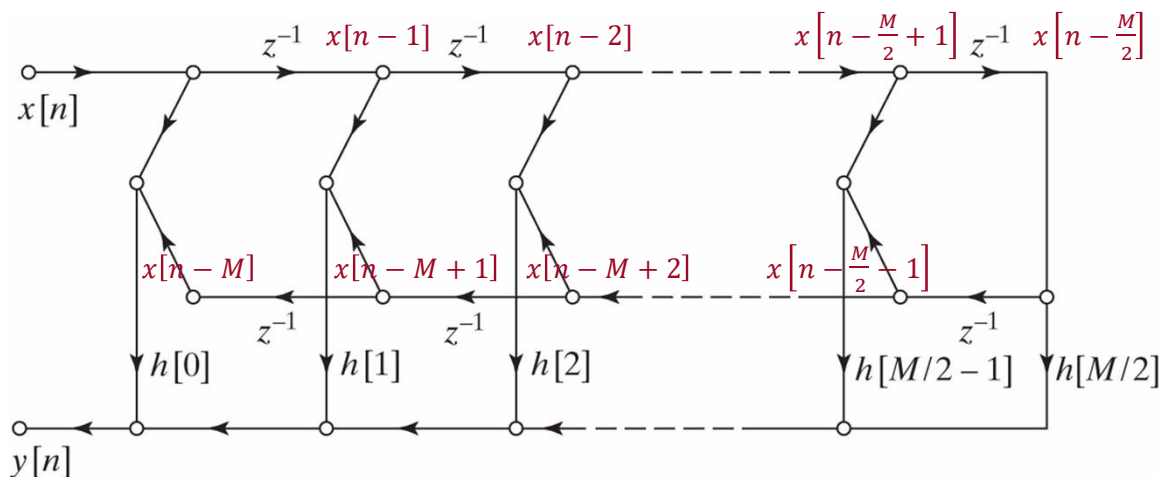
STRUCTURES FOR LINEAR-PHASE FIR SYSTEMS

- Generalized linear-phase FIR systems have symmetric impulse responses

$$h[M - n] = h[n] \quad \text{or} \quad h[M - n] = -h[n] \quad \text{for } n = 0, 1, \dots, M$$

- Due to this symmetry, the number of multipliers can be essentially halved
- For example, if $h[M - n] = h[n]$ and M is even, then ([see notes](#))

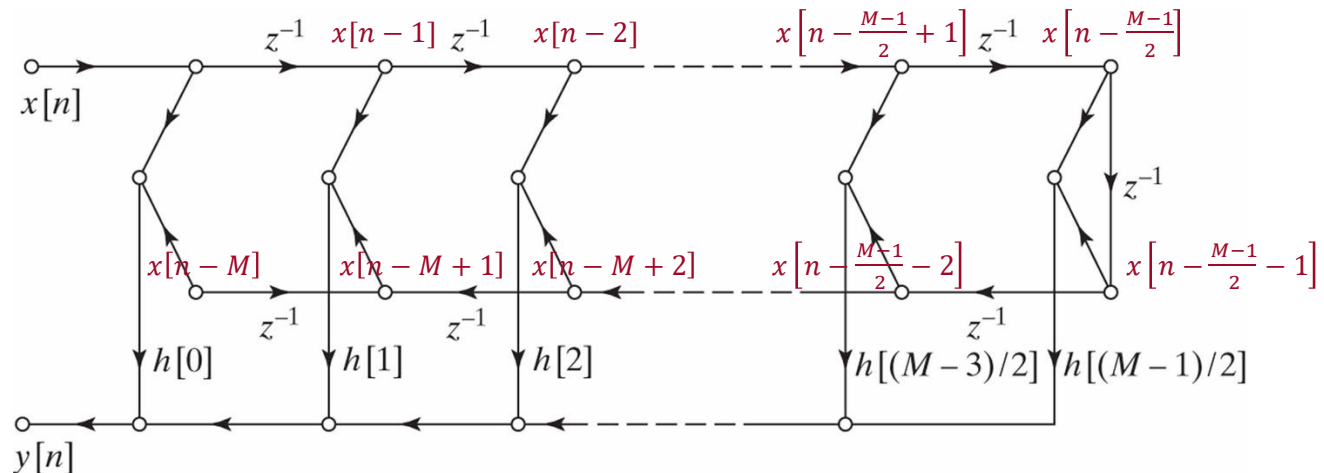
$$y[n] = \sum_{k=0}^{M/2-1} h[k](x[n - k] + x[n - M + k]) + h\left[\frac{M}{2}\right] x\left[n - \frac{M}{2}\right]$$



STRUCTURES FOR LINEAR-PHASE FIR SYSTEMS

- On the other hand, if $h[M - n] = h[n]$ and M is odd, then ([see notes](#))

$$y[n] = \sum_{k=0}^{(M-1)/2} h[k](x[n - k] + x[n - M + k])$$



READING ASSIGNMENT

- Read Sections 6.1 – 6.5 in the textbook
- Start reading Chapter 7